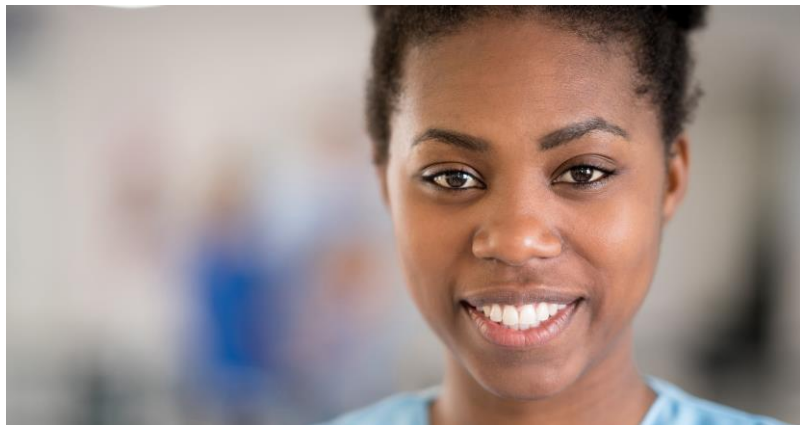
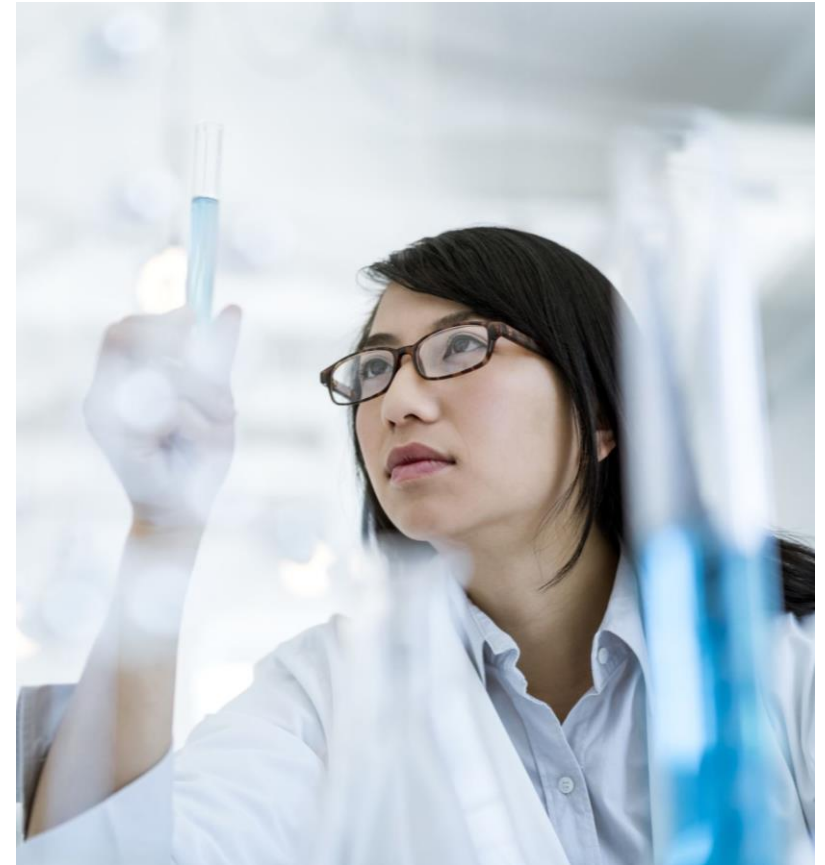




PROJECT 4.5 — Mixed Analytics

MetroCare NHS Trust — Extended Operational Insights Project



Project Overview

MetroCare NHS Trust is continuing to experience pressure on patient flow, appointment attendance, and financial performance. As a Junior Data Analyst, you have been asked to extend your work from Project 4.4 by analysing **two new datasets**:

- **Appointments**
- **Billing**

Your goal is to produce a short analytical report supported by SQL queries and a compact Power BI dashboard.



Datasets Provided

You will receive **six datasets**:

1. Patients
2. Admissions
3. Departments
4. StaffingLevels
- 5. Appointments (NEW)**
- 6. Billing (NEW)**



Tasks

1. SQL Analysis (5 Queries)

Write and submit **five SQL queries**:

1. Join **Admissions + Patients** to show patient demographics for each admission.
2. Join **Appointments + Departments** to count appointments by department.
3. Calculate the **No-Show Rate** (percentage of appointments with Status = 'No-Show').
4. Summarise **Billing** to show total charges per patient.
5. Create a **multi-join query** (Admissions + Billing + Patients) showing total cost per admission with patient details.

Submit your SQL as a `.txt` or `.sql` file.



Tasks

2. Power BI Dashboard

Page 1

Page 1 — Operational Overview

Include:

- Admissions by month
- Appointment attendance
(Completed vs No-Show)
- Staffing levels vs admissions
- Department slicer



Tasks

2. Power BI Dashboard

Page 2

- **Page 2 — Financial Overview**
- Include:
 - Total billing by department
 - Average charge per patient
 - Billing trend over time
- **Required DAX Measures**
- You must create at least **four**:
 - Total Admissions
 - No Show Rate
 - Total Billing
 - Average Charge Per Patient



Tasks

3. Written Summary

Write a **150–200 word** analytical summary covering:

- Key operational findings
- Appointment attendance patterns
- Financial insights
- Any notable trends or anomalies



Tasks

4.

Recommendations

Provide **two evidence-based recommendations** for improving:

- Patient flow
- Appointment attendance
- Financial efficiency



Tasks

5. Visual Exports

Export **two visuals** (one from each page) as PNG.

